



SCSM2223-CROSS-PLATFORM APPLICATION DEVELOPMENT

SEMESTER II, SESSION 2025/2026

SECTION 04

CHAPTER 8 — INDIVIDUAL ASSIGNMENT

REPORT

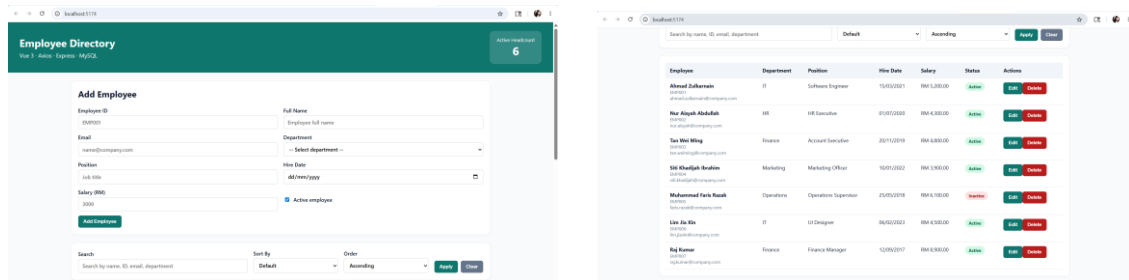
LECTURER: DR. HISHAMMUDDIN BIN ASMUNI

Github: https://github.com/yuanman1021/CPAD_Assignment1

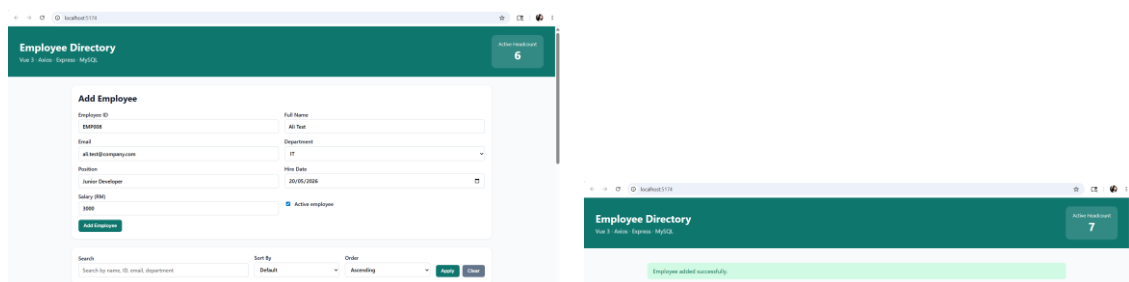
NAME	MATRIC NUMBER
KHOO YUAN MAN	A24CS0096

1.0 Screenshots

1. Main Employee Directory page



2. Add employee form



(Fill the table)

(Successful message)

3. Employee table

Employee	Department	Position	Hire Date	Salary	Status	Actions
Ahmad Zulkarnain EMP001 ahmad.zulkarnain@company.com	IT	Software Engineer	15/03/2021	RM 5,200.00	Active	Edit Delete
Nur Aiyah Abdullah EMP002 nur.aiyah@company.com	HR	HR Executive	01/07/2020	RM 4,300.00	Active	Edit Delete
Tan Wei Ming EMP003 tan.wei@company.com	Finance	Account Executive	20/11/2019	RM 4,800.00	Active	Edit Delete
Siti Khadijah Ibrahim EMP004 siti.khadijah@company.com	Marketing	Marketing Officer	10/01/2022	RM 3,900.00	Active	Edit Delete
Muhammad Faris Razak EMP005 faris.razak@company.com	Operations	Operations Supervisor	25/05/2018	RM 6,100.00	Inactive	Edit Delete
Lim Jia Xin EMP006 lim.jiaxin@company.com	IT	UI Designer	06/02/2023	RM 4,500.00	Active	Edit Delete
Raj Kumar EMP007 raj.kumar@company.com	Finance	Finance Manager	12/09/2017	RM 8,900.00	Active	Edit Delete

(Updated data after adding new employee)

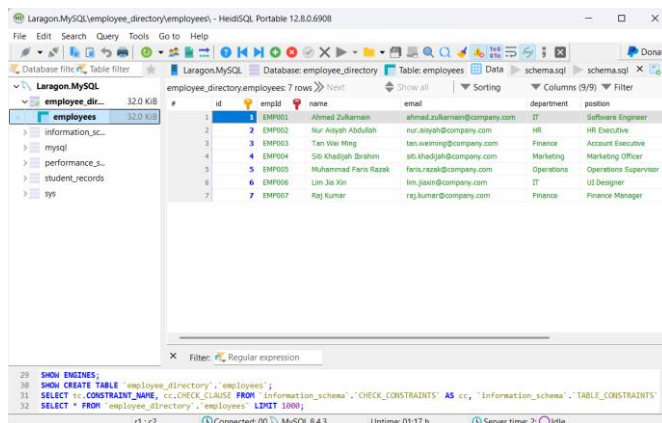
4. Search result

Search: Finance Sort By: Default Order: Ascending Apply Clear

Employee	Department	Position	Hire Date	Salary	Status	Actions
Tan Wei Ming EMP003 tan.wei@company.com	Finance	Account Executive	20/11/2019	RM 4,800.00	Active	Edit Delete
Raj Kumar EMP007 raj.kumar@company.com	Finance	Finance Manager	12/09/2017	RM 8,900.00	Active	Edit Delete

(Search Finance, only Finance employees appear.)

8. MySQL table in HeidiSQL



#	id	empId	name	email	department	position
1	EMP001	1	Almad Zulkarnain	almazul@company.com	IT	Software Engineer
2	EMP002	2	Nur Anisah Abdullah	nur.anisah@company.com	HR	HR Executive
3	EMP003	3	Tan Wei Ming	tan.wei@company.com	Finance	Account Executive
4	EMP004	4	Siti Khadijah Ibrahim	siti.khadijah@company.com	Marketing	Marketing Officer
5	EMP005	5	Muhammad Fariq Razali	fariq.razali@company.com	Operations	Operations Supervisor
6	EMP006	6	Lim Ja Xin	lim.jaxin@company.com	IT	UI Designer
7	EMP007	7	Raj Kumar	raj.kumar@company.com	Finance	Finance Manager

(After load data, run, refresh, then view data table of employee directory.)

2.0 Learning Outcomes

LO1: Connect Vue to Backend

The Vue frontend connects to the Express backend using Axios. The backend runs on port 3001 and the frontend runs on port 5174.

LO2: Axios HTTP Client

The project uses a single Axios instance in `src/api/employeeApi.js`. It includes a `baseUrl`, `timeout`, `default headers`, `request interceptor` and `response interceptor`.

LO3: Form Handling in Vue

The `EmployeeForm.vue` component uses `v-model`, `.trim` and `.number` modifiers. It validates employee ID, name, email, department, position, hire date and salary before submitting.

LO4: Persistent CRUD

The Express API supports POST, GET, PUT and DELETE endpoints. All changes are saved in the MySQL employees table using prepared statements.

3.0 Challenges and Solutions

One of the challenges I faced was the Vue frontend could not load employee data because the frontend and backend were running on different ports. To solve this, I made sure the Express backend runs on `http://localhost:3001` and the Vue frontend runs on `http://localhost:5174`. I also used `cors()` in `server/index.js` so that the Vue app can send Axios requests to the backend.